

BELOGORSK, Russian Federation

WMO: 315130

Lat: 50.917N

Lon: 128.467E

Elev: 583

StdP: 14.39

Time Zone: 9.00 (E09)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.8	-25.3	-32.3	0.9	-28.2	-29.2	1.1	-24.7	11.0	1.6	9.6	0.6	1.4	320	0.529

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.4	87.4	69.7	84.3	69.0	81.3	68.1	73.3	82.2	71.8	80.2	70.3	78.4	5.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.6	115.0	77.6	69.0	109.0	76.2	67.5	103.4	75.1	37.5	83.0	36.0	80.0	34.8	78.4	85.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.0	12.2	10.4	DB	-32.6	93.7	5.7	4.9	-36.7	97.2	-40.1	100.1	-43.3	102.8	-47.4	106.4	(4)
(5)			WB	-32.7	77.5	5.6	3.5	-36.8	80.0	-40.1	82.1	-43.3	84.0	-47.3	86.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	33.4	-9.5	-1.0	16.9	38.7	54.7	66.6	71.5	67.6	54.5	35.9	11.2	-7.9
	DBStd	30.30	8.17	8.18	11.30	8.80	7.55	6.70	4.67	5.60	8.15	9.07	11.90	8.76
	HDD50	8139	1845	1429	1025	353	37	0	0	1	47	444	1163	1795
	HDD65	11966	2310	1849	1490	788	331	57	6	37	323	902	1613	2260
	CDD50	2095	0	0	0	15	182	498	665	545	184	6	0	0
	CDD65	447	0	0	0	0	11	105	206	116	9	0	0	0
	CDH74	3420	0	0	0	12	174	995	1413	747	79	0	0	0
	CDH80	939	0	0	0	2	35	370	379	146	7	0	0	0
Wind	WSAvg	4.3	3.1	3.7	4.8	6.3	5.7	4.2	4.0	4.0	4.4	4.5	4.0	3.5
Precipitation	PrecAvg	21.5	0.3	0.3	0.4	0.9	2.2	3.1	4.8	4.5	2.6	1.2	0.6	0.5
	PrecMax	35.0	1.0	1.3	1.6	3.5	5.2	6.2	9.9	13.0	7.3	3.3	1.3	1.3
	PrecMin	14.2	0.0	0.0	0.0	0.0	0.3	0.8	1.7	0.9	0.7	0.4	0.0	0.0
	PrecStd	4.3	0.2	0.2	0.4	0.8	1.3	1.3	2.2	2.4	1.6	0.7	0.4	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.7	26.2	49.3	74.6	84.7	96.5	92.2	87.0	80.4	65.1	43.8	20.4
		MCWB	12.7	24.0	38.7	53.0	60.6	73.1	71.0	71.7	65.7	51.1	38.8	19.0
	2%	DB	9.4	21.6	43.7	64.5	79.1	88.2	87.2	83.9	76.1	58.1	38.0	13.0
		MCWB	7.8	18.8	34.9	48.1	57.9	67.9	72.2	70.5	62.6	47.1	32.7	11.5
	5%	DB	5.8	16.4	39.0	58.6	74.1	84.3	84.0	81.2	72.2	53.3	33.0	9.0
		MCWB	4.3	14.1	31.8	45.1	55.8	66.3	70.7	69.3	60.7	44.7	29.5	7.7
	10%	DB	2.4	11.9	34.2	53.7	69.6	79.4	81.2	78.3	68.0	49.6	29.0	5.2
		MCWB	1.3	9.9	28.9	42.0	53.7	65.5	69.7	67.5	58.6	42.3	25.9	4.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.6	24.3	39.1	54.7	62.2	73.2	75.6	75.1	67.5	53.1	38.2	18.9
		MCDB	14.5	25.3	47.7	69.4	78.8	89.9	84.2	82.9	75.8	59.3	42.8	20.2
	2%	WB	8.0	18.8	35.4	49.6	59.7	70.2	73.8	72.8	64.9	49.2	33.1	11.7
		MCDB	9.3	21.4	42.5	63.2	74.2	83.6	82.3	80.7	72.6	55.3	37.1	12.7
	5%	WB	4.3	14.3	31.9	45.8	57.3	68.5	72.4	70.9	62.4	45.6	29.9	7.7
		MCDB	5.5	16.4	38.3	57.2	70.3	80.9	80.9	78.2	69.6	52.3	33.1	9.0
	10%	WB	1.3	10.0	29.3	42.7	55.3	66.6	71.2	69.3	60.2	42.4	25.7	4.1
		MCDB	2.3	11.8	34.1	52.2	67.1	76.6	79.3	75.9	67.1	48.8	28.8	5.1
Mean Daily Temperature Range	5% DB	MDBR	16.0	19.1	19.4	19.0	20.1	17.7	16.4	17.1	19.4	17.1	15.5	14.5
		MCDBR	16.7	18.7	20.9	27.1	27.6	25.4	20.6	21.2	24.4	23.2	16.5	15.3
		MCWBR	15.3	16.6	13.1	14.9	12.1	9.4	8.2	9.5	11.8	13.8	12.2	14.0
	5% WB	MCDBR	16.7	17.8	20.0	25.3	22.7	21.7	17.0	17.0	20.6	19.2	15.5	14.9
		MCWBR	15.3	15.9	12.9	14.3	10.9	9.6	7.9	8.7	10.5	13.0	11.6	13.7
Clear-Sky Solar Irradiance	taub	0.268	0.301	0.380	0.543	0.599	0.498	0.482	0.429	0.396	0.397	0.323	0.272	
	taud	2.134	2.060	1.893	1.692	1.657	2.003	2.099	2.238	2.242	2.063	2.096	2.085	
	Ebn at Noon	235	256	251	221	216	244	246	254	248	214	204	208	
	Edh at Noon	24	34	50	69	75	53	48	39	35	35	25	22	
All-Sky Solar Radiation	RadAvg	512	900	1404	1602	1726	1889	1748	1526	1201	786	505	396	
	RadStd	15	21	79	168	135	222	130	99	104	40	27	21	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (14 neighbors)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page